Owen Fox

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Education

Villanova University — Villanova, PA
Bachelor of Science in Computer Science — *August 2022 - May 2026*

- **GPA**: 3.70 / 4.00; **Honors:** Cum Laude, Dean's List (7 Semesters)

Projects

Bullpen Engine | *TypeScript, Next.js, PostgreSQL, Prisma, Docker*

- Developed full-stack TypeScript application with Next.js App Router, implementing RESTful APIs for real-time simulation execution and PostgreSQL database integration via Prisma ORM
- Engineered pitcher performance model sampling K/9, BB/9, HR/9 rates to generate plate appearance outcomes, advancing baserunners through statistically accurate win probability calculations
- Containerized application with Docker Compose for single-command deployment, including database seeding with 30 MLB reliever profiles and automated schema migration

PokeMahn | *Godot, `C#`*

- Developed PokeMahn, a Pokémon-inspired RPG built in the Godot game engine with `C#`, featuring a tilemap overworld, animated player movement, encounter triggers, and battle logic
- Implemented core gameplay mechanics, including map navigation, collision detection, and encounter systems to simulate exploration and turn-based battle triggers
- Structured game logic using Godot's scene and node system, modularizing components for environment, player control, UI messaging, and battle events

Space Invaders | *Python*

- Developed a fully interactive Space Invaders game in Python using the Turtle graphics library, implementing real-time rendering, keyboard input handling, and a custom game loop running around 60 FPS
- Managed game state and HUD rendering, tracking score, game-over conditions, and user feedback through on-screen updates and event-driven callbacks
- Designed modular game systems including player movement, projectile mechanics with cooldown logic, alien formation movement with edge bouncing and timed descent, and win/lose state management

Experience and Leadership

AI Trainer – Software Engineering / Coding Domain — Remote
Handshake AI Fellowship (Contract) — *June 2026 – October 2026*

- Evaluated and scored responses from frontier large language models (LLMs) on multi-turn software-engineering and coding tasks, judging response quality, helpfulness, and communication
- Applied a structured evaluation rubric to rate model completions on a 1–7 scale across dimensions including helpfulness, technical clarity, tone and communication style, response structure, and how well the model adapts to and collaborates with the user
- Authored detailed written rationales for each score and performed comparative (preference) evaluations across multiple candidate responses to the same coding prompt
- Maintained high inter-rater agreement and scoring consistency, passing a rigorous calibration screening before contributing to production data
- Generated high-quality human-feedback data used to improve the reasoning, communication, and overall response quality of AI coding assistants

Vice President — Villanova, PA
Villanova Software Engineers — *October 2024 – May 2026*

- Served as Vice President of a 20+ member student engineering organization building full-stack applications for startup clients, leading frontend development and coordinating closely with backend, design, and client-facing teams
- Mentored 10+ junior members through weekly code reviews and collaborative projects, reinforcing best practices in Git workflows, debugging, and frontend architecture to improve code quality and accelerate onboarding

Competitive Programmer — Villanova, PA
Villanova Programming Team — *August 2022 – May 2026*

- Solved 200+ algorithmic challenges on LeetCode and Kattis through weekly practice sessions, strengthening competitive programming performance and advancing mastery in algorithmic strategies, dynamic programming, and complexity analysis
- Collaborated in a 3-member programming team to strategize and optimize real-time problem-solving workflows for competitive events, improving coordination, time management, and debugging efficiency across high-pressure contests

Technical Skills & Certifications

Languages: JavaScript, TypeScript, Python, Java, `C#`, HTML, CSS
Frameworks & Libraries: React, Next.js, FastAPI, TailwindCSS, Vite, Prisma, SwiftUI
Tools & Platforms: GitHub, Docker, Firebase, Jira, Vercel, Xcode
Certifications: LinkedIn Python and Java Skill Assessments